Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/25/2020
PI Name:	Liu, Bin Ph.D.		
Project Title:	Three-Dimensional Dusty Plasma Experiments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bin-liu@uiowa.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	319-335-1686
Organization Name:	University of Iowa		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	203 Van Allen Hall		
PI Web Page:			
City:	Iowa City	**State:**	IA
Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	07/25/2017	**End Date:**	04/25/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goree, John Ph.D. (University of Iowa)		
Grant/Contract No.:	JPL 1579454		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A flight experiment will be performed to study phenomena resulting from shear motion in a dusty plasma. Applying an external shear, we will probe fundamental material properties selected from these topics: lattice wave excitation and coupling, viscous transport, and microscopic motion under frictional conditions. We will take a microscopic approach to uncover the nature of macroscopic properties. We will use mesoscopic particles, which serve as analogs of molecules, but allow tracking of individual particles as they interact. The experiment will use dusty plasmas, also known as "complex plasmas." These are four-component mixtures of micron-size solid particles, electrons, ions, and neutral gas. The solid particles obtain large electric charges, so that they interact by Coulomb forces with neighboring particles. The charges are so large that these particles behave collectively as a strongly-coupled plasma. The cloud of solid particles constitutes the sample that is studied. Microgravity enables our experiment with a three-dimensional sample. Without microgravity, under 1g conditions, sedimentation causes the dust cloud to settle into a 2D layer, or at most a few horizontal layers. Our microgravity experiment will extend the understanding of particle motion that was gained in previous ground-based experiments with smaller 2D samples. Our flight experiment will be performed using Plasma Kristall–4 (PK-4), which is a facility in the Columbus module of the International Space Station (ISS). It was built by the European Space Agency. The main diagnostic is video imaging using cameras. Image data will be analyzed to generate particle position and velocity. Physical measures that can be obtained from these image data, using methods developed for ground-based experiments, include a wave's propagation speed and dispersion relation, temperature profile, flow velocity, or shear stress. SCIENTIFIC/TECHNICAL MERITS The experiment's goals are to answer at least one of these fundamental physics questions: (1) Can we exploit the unique properties of dusty plasma to verify the theoretical prediction of localized viscous heating in a 3D liquid? (2) Can hydrodynamic shear motion couple into the generation of longitudinal waves? (3) What is the dispersion relation of lattice waves in a liquid? (4) Does a three-dimensional dusty plasma exhibit stick-slip motion during plastic deformation? The experiment will exploit manipulation devices on PK-4 including laser manipulation. Laser light imparts momentum to the solid particles, causing them to flow through the surrounding sample region. This modulation will be a pulse that is either long or short, according to the scientific goal. The feasibility of our flight experiment is based on previous ground-based experiments and our simulation. Our laboratory experiments have confirmed localized viscous heating and lattice waves. Ground-based experiments by other scientists have also confirmed the existences of plastic flow conditions and stick-slip motion. Using a simulation under PK-4 conditions, we have established the physical mechanism of wave excitation by a pulsed shear flow. In the unlikely event that this flight experiment does not yield useful results, we have a plan for a backup experiment to investigate the superheating and supercooling phase transitions, or non-Gaussian statistics. BROADER IMPACTS This research has an overlap with statistical physics, fluid physics, and soft condensed matter physics, as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, and a website. A K-12 outreach presentation will be made at an elementary school.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has an overlap with statistical physics, fluid physics, and soft condensed matter physics, as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, and a website. A K-12 outreach presentation will be made at an elementary school.
	Our accomplishment during this reporting period includes our first flight experiment on the International Space Station (ISS), investigating pulsed shear motion in three-dimensional (3D) dusty plasmas in the Plasma Kristall–4 (PK-4). We planned and successfully executed this experiment in PK-4 Science Campaign 7. Progress has also been made in analyzing the data from this flight experiment, including particle tracking, new analysis method development, structural and dynamical characterization of the 3D dusty plasmas under microgravity. The goal of this research is to gain a fundamental understanding of the microscopic and macroscopic properties of 3D physical systems, at the particle level. We focus on the dusty plasmas that are strongly coupled, where the dust component behaves collectively like a solid or a liquid. Our objective is to investigate how these dusty plasmas respond to a pulsed shear, at a small (atomic-like) time scale. Direct imaging of dust motion offers a unique opportunity for us to answer this fundamental physics question. ISS FLIGHT EXPERIMENT Our first ISS flight experiment on three-dimensional (3D) dusty plasmas was successfully completed in PK-4 Science Campaign 7, July 2019. The primary goal of this flight experiment was to make a pulsed flow of dust particles using the PK-4 manipulation laser. Key requirements included the formation of a stable dust cloud that is strongly coupled and the implementation of two schemes to modulate the manipulation laser. We verified the feasibility of our experiment design. We also explored parameter regime to find conditions suitable for observing the evolution of shear motion as the shearing force applied by the manipulation laser is varied. Below are some major achievements in this experiment: • making a stable dust cloud that resembles solids or cold liquids, • testing two different schemes for modulating the manipulation laser, • observation of the structure evolution in response to pulsed and step-wise shearing forces, • testing the effect of gas pressure. Although this flight experiment met its goals, we also identified an issue that needs to be addressed in a future experiment. The dust cloud exhibited a small drift, which would complicate our data analysis in extracting the shear motion. This can be easily solved by adjusting the duty cycles of the voltage pulses that sustain our plasma. We concluded that another ISS flight experiment is much needed for our shear motion project. We will need to address

	the minor issue identified above, and we will also need to narrow our parameter range, for example, for the manipulation laser power. More experiment time is also required for recording adequate data for future publications. Principal Invesitigator (PI) Bin Liu participated in the execution of this experiment in Toulouse France, together with PK-4 Core-Team scientists. He also observed other PK-4 flight experiments performed in Campaign 7. To prepare for this flight experiment, we worked closely with PK-4 Core Team scientists. At Iowa, we designed our experiment procedure, configuration, and parameters regime, yielding a flow chart. PK-4 Core Team generated an experiment script based on our flow chart. PI Bin Liu travelled to the DLR (German Space Agency), and tested the script using the PK-4 ground model at the DLR.
Task Progress:	ANALYSIS OF ISS PK-4 FLIGHT DATA During this report period, we have performed preliminary analysis of the ISS flight data from our shear motion experiment. For our flight data, which are primarily 2D images of a slab of 3D dust cloud, one of the most crucial steps is to track particles over frames. As compared with 2D dusty plasmas, particle tracking in 3D is much more challenging and requires to be more careful. We have tested our flight data for particle tracking, finding that about 70% of identified particles can be tracked for two consecutive frames. However, further effort is required to verify the accuracy of the processed data. Other achievements in data analysis include: • Structural characterization. Using a method that we developed previously, we characterized the structure of our 3D dust cloud and obtained a 3D pair correlation using 2D image data. We found that our dust cloud had a structure like a solid, with a particle spacing of about 0.35 mm. We also determined the number density of the 3D dust cloud, which was about 3.3E-10 m^-3. • New analysis method development. In order to characterize the shear motion and extract dust-fluid velocity without tracking individual particles, we are developing a new method that centers on the space-time diagram of image brightness. We have obtained some preliminary results, which will be compared with that from tracking individual particles. • Collective motion. Using particle position, we calculated a microscopic particle density, yielding a spectrum of particle motion. This spectrum is a result of the decomposition of particle natural motion into collective modes with different wave numbers. PK-4 FLIGHT DATA During this report period, the Iowa group received a portion of the flight data from Science Campaigns 7, which was from our ISS flight experiment for pulsed shear motion. FLIGHT EXPERIMENT PLANNING Our planning involved a collaboration with other institutions or agencies. Significant collaborators included PK-4 Core Team members in Germany and Russia, PK-4 Facility Science Team (FST), ESA (European Space Agency) officials, CADMOS (Microgravity and Spatial Operations Development Support Center) at the CNES (France National Center for Space Studies), and the collaborators specific to Iowa projects. PI Bin Liu interacted with these collaborators frequently via email and occasionally by phone, and we also met them at various meetings and conferences. One significant meeting was the recent PK-4 meeting at the DLR (German Space Agency) in December 2019. PI Bin Liu attended this meeting, and he gave a talk regarding the progress of Iowa PK-4 projects.

Bibliography Type:	Description: (Last Updated: 10/05/2022)
Abstracts for Journals and Proceedings	Goree J, Liu B. "Dusty plasma experiments under microgravity conditions." Presented at the 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Liu B, Goree J, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Molotkov VI, Petrov OF, Thoma MH. "Nonlinear dust acoustic waves in a plasma under microgravity conditions." Presented at the 61st Annual Meeting of the American Physical Society Division of Plasma Physics, Fort Lauderdale, Florida, October 21-25, 2019. Bulletin of the American Physical Society. 2019;64(11):NP10.00075. http://meetings.aps.org/Meeting/DPP19/Session/NP10.75 , Oct-2019
Abstracts for Journals and Proceedings	Mir AA, Tiwari S, Sen A, Ganguli G, Crabtree C, Liu B, Goree J. "Beat waves in dusty plasma." Presented at the 12th International Conference on Plasma Science and Applications (ICPSA), Lucknow, India, November 11-14, 2019. Abstracts. 12th International Conference on Plasma Science and Applications (ICPSA), Lucknow, India, November 11-14, 2019. , Nov-2019